# Behavior Service Guide

## Description

When a user interact with Alpha Mini, it can feedback with various behaviors. The behavior is defined by different scenarios, which is usually related to voice command; for example, after the user said “Good morning”, Alpha Mini will smile and wave the left hand, also with greeting words to respond to the user. Meanwhile, the LED light, as the mouth of Alpha Mini will flash, just like the robot is talking! Apart of voice command, there are other scenarios might trigger behaviors, like system error, fall-down or tap on head.

A behavior consists of action, expression, sound and light, which calls motion service, expression service, speech service and light service of the robot system. This document indicates basic structure and field of a behavior configuration file which defines parameters and sequences to call robot system service.

A behavior configuration file is an embedded XML in robot system directory. In official version, those files are packed into ROM with robot apk and stored under **"/system/files/behaviors/"** directory; for debug version, behavior configuration files are stored under **"/sdcard/behaviors"** directory after pushed into the robot via ADB command

## Behavior Configuration File

There are 5 kinds of Alpha-Mini behaviors can be configured: *ACTION, EXPRESS, TTS, LED and MUSIC.* In the xml, the behavior can be configured as:

### Public Field

| Field | Type | Description |
|---|---|---|
| seqNo | int | >=1, set the sequence of the behavior, default 1, not compulsory |
| delayTime | int | Delay how many MS to act the behavior, default 0, not compulsory |
| loopCount | int | Repeat the behavior for how many times, default 1, not compulsory |

*Once the file is complied, the robot will first start the behavior with the minimum **seqNo**, when all the fields under some **seqNo** are acted, the robot will act field with the next **seqNo**.

### Action

| Field | Type | Description |
|---|---|---|
| name | String | Name of the action file (without file extension), compulsory |

* action file is a **.ubx** file，which is made by "Action Configuration Tool" and stored under "**/system/files/actions/**" directory (or under "**sdcard/actions/**" for debug version)

### Express

| Field | Type | Description |
|---|---|---|
| name | String | Name of the expression file (without file extension), compulsory |

* Expression file is a **.gif** or **.json** file，which is made by "AE" or other Animation edition tools. The size of the animations support **240x240, 480x240**, could be gif format or lottie animation. The expression files are stored under "**/system/files/express/**" directory (or under "**sdcard/express/**" for debug version)

**TTS**

| Field | Type | Description |
|---|---|---|
| text | String | Name of the expression file (without file extension), compulsory |

*If the TTS field is related to file:///***.mp3** file，which is a local TTS file made from TTS provider. The local TTS files are stored under "**/system/files/localTts/**" directory (or under "**sdcard/ localTts /**" for debug version); if the TTS field is normal text content, please connect with the online engine.

**LED**

| Field | Type | Description |
|---|---|---|
| effect | int | 1~4: 1.Always on 2. Breathe 3. Flash 4.RGB Switch, compulsory |
| color | String | Color for the light, hexadecimal number (e.g #FF2430), compulsory |
| runTime | int | >0, the light effects for how many MS, compulsory |

## MUSIC

| Field | Type | Description |
|---|---|---|
| url | String | URL address of the music (file or web link), compulsory |

*If the Music field is related to file:///***.mp3** file，which are stored under "**/system/files/music/**" directory (or under "**sdcard/ music /**" for debug version); If the Music field is related to URL address, please connect with the online music provider.

**Set**

Set tag contains a group of the behaviors, also it can define seqNo, delayTime and loopCount of the behavior. The set is finished only after all the behaviors are finished, which can be used for control the running sequence of a behavior group. The set tag is nested, which is a very important to implement complicated behaviors.

Sample Code

```
<?xml version="1.0" encoding="utf-8"?>
<set>
    <tts seqNo="1" delayTime="20" loopCount="1" text="Hey,  this is the first dance"/>
    <led seqNo="1" delayTime="20" loopCount="1" effect="3" color="#FF00FF"
runTime="3000" />
    <express seqNo="1" name="emo_013" delayTime="10" loopCount="1" />
    <action seqNo="2" name="dance_0001" delayTime="10" loopCount="1" />
    <music seqNo="2" name="file:///dance_0001.mp3" delayTime="10"
loopCount="1" />
    <set delayTime="30" seqNo="3" loopCount="1">
        <tts seqNo="1" delayTime="20"text=" Hey,  this is the second dance " />
        <led seqNo="1" delayTime="0" effect="3" loopCount="1" runTime="3000"
color="#ff0023" />
        <express seqNo="1" name="emo_013" delayTime="0" loopCount="1" />
        <express seqNo="2" name="emo_014" delayTime="0" loopCount="5" />
        <express seqNo="3" name="emo_015" delayTime="0" loopCount="10" />
        <action seqNo="2" name="dance_0002" delayTime="20" loopCount="1" />
        <music seqNo="2" name="file:///dance_0002.mp3" delayTime="20"
loopCount="1" />
    </set>
</set>
```

## Creation and Execution Behaviors in java

### Create A Behavior by Loading xml File

```
    class demo {

        public void executeBehaviorFromFile() {

        Behavior behavior =

BehaviorInflater.loadBehaviorFromXml(behavior_file);

        behavior.setPriority(Priority.NORMAL);

        behavior.start();
```

```
        behavior.setBehaviorListener(new Behavior.BehaviorListener() {
            @Override public void onCompleted(BehaviorStopCaused caused) {
                Log.w("logic", "behavior completed...");
                }
            });
        }
    }
```

**Create A Behavior by Builder**

```
public void executeBehavior() {
    BehaviorSet.Build builder = new
BehaviorSet.Builder().setOrder(1).setLoopCount(1).setDelayTime(0);
    builder.addBehavior(new ActionBehavior.Builder("action-name").build());
    builder.addBehavior(new ExpressBehavior.Builder("express-name").build());
    Behavior behavior = builder.build();
    behavior.setBehaviorListener(new Behavior.BehaviorListener() {
        @Override public void onCompleted(BehaviorStopCaused caused) {
            Log.w("logic", "behavior completed...");
        }
    });
    behavior.start();
    }
}
```

**BehaviorStopCaused Description**

A group of Behavior is usually with an ActionBehavior. The ActionBehavior should call Action Service to act actions, and the actions could be interrupted: For example, servos in protection mode (when a servo is overcurrent or overvoltage caused by external factors, the servo will loss the power supply, as the protection mode, to prevent further damage) or an action with higher priority is ongoing (it causes the conflicts if the same servos are called). In this case, the ActionBehavior will be stopped and stop the whole BehaviorSet (notice: if other behaviors like *EXPRESS, MUSIC, TTS or LED* is stopped, it will not effect on whole BehaviorSet). By listen to BehaviorListener, function BehaviorStopCaused can be called and get the reason why the BehaviorSet is stopped.

```
//### BehaviorStopCaused
//The BehaviorSet is completed
public boolean isStoppedByCompleted() {
return cause == 0;
}
//The BehaviorSet is forced to stop by havior.stop()
public boolean isStoppedBySelf() {
return cause == 1;
}
// Action in the BehaviorSet is stopped by ActionApi.stopAction()
public boolean isInturreptedByStopAction() {
return cause == 10001;
}
// The BehaviorSet has received motor broken and is stopped
public boolean isInturreptedByMotorBroken() {
return cause == 10002;
}
// The BehaviorSet has received motor in protection and is stopped
public boolean isInturreptedByMotorProtected() {
return cause == 10003;
}
```

```
// Action in the BehaviorSet is stopped by ActionApi.playAction() with higher priority
public boolean isInturreptedByOtherAction() {
return cause == 10004;
}
// Action in the BehaviorSet cannot be found
public boolean isStopppedByActionFileNotFound() {
return cause == 10005;
}
// Action in the BehaviorSet is stopped by error
public boolean isStoppedByError() {
return cause != 0
&& cause != 1
&& cause != 10001
&& cause != 10002
&& cause != 10003
&& cause != 10004;
}
```

**Notice:**

1. Please refer to sdk-demo project to know more details.

2. behavior-lib.jar has nested in on Alpha-Mini robot-sdk.jar.